

* Bio-fuel

Any fuels that is derived from bio-mass that is plant, algae or animal waste, is called biofuel. Biofuels is considered to be renewable energy, unlike fossil fuels such as petroleum, coal and natural gas.

→ Classification of Biofuels

1. First Generation Biofuels:

Biofuels made from food crops.

2. Second Generation Biofuels:

Biofuels made of ligno-cellulose materials (biomass).

3. Third Generation Biofuels:

Biofuels made of agricultural waste- or waste plant materials. Biofuels from fast growing non-food crops (Jatropha).

4. Fourth Generation Biofuels:

Biofuels from genetically modify energy crops.

→ Most common biofuels are; Ethanol, bio-diesel and bio-gas.

→ The liquid biofuels are ethanol and biodiesel.

[Ethanol]

→ Ethanol is a type of alcohol that can be produced

using any feed-stocks which contains significant amount of sugarcane, sugarbeet and starch, etc.

→ Bio-ethanol is used to generate fast motor power (transportation and electricity).

[Diesel]

→ Bio-diesel is renewable, bio-degradable fuel manufactured from vegetable oils and animal fats.

→ Trans-Esterification is a chemical process which converts natural ~~oils~~ fats & oils into acid and alcohol.

→ The most common bio-diesels are; B-5 [5% biodiesel & 95% diesel] and B-20 [20% biodiesel & 80% diesel].

→ Bio-diesel ~~raises~~ cetane number of fuel.

→ Bio-diesel are clean and green fuel, it improves the fuel lubricity.

→ Advantages of Biofuel

- Energy efficiency
- Reducing foreign oil dependency.
- Health benefits.
- Positive economic impact.
- Reducing green house gases.
- Sustainability.
- High quality engine performance.